In [4]:
```
# Import niezbędnych bibliotek i ustawienie wykresów
import pandas as pd
import matplotlib.pyplot as plt
%matplotlib inline

# Tworzenie ramki danych ze słownika
slownik = {
    'Kraj': ['Polska', 'Niemcy', 'Czechy'],
    'Odsetek_palacych': [15.2, 12.1, 14.5],
    'Rok': [2019, 2019, 2019]
}
df_ze_slownika = pd.DataFrame(slownik)
print("--- Ramka ze słownika ---")
display(df_ze_slownika)

# Zachowanie ramki danych pobranych z pliku CSV
nazwa_pliku = 'IHME_GBD_2019_SMOKING_AGE_1990_2019_PREV_Y2021M05D27.CSV'
df = pd.read_csv(nazwa_pliku)
print("\n--- Pomyślnie wczytano dane z pliku CSV ---")

# Tworzenie ramki danych z listy list
lista_list = [
    ['Francja', 13.0, 2019],
    ['Hiszpania', 16.2, 2019],
    ['Włochy', 14.8, 2019]
]
df_z_listy = pd.DataFrame(lista_list, columns=['Kraj', 'Odsetek_palacych', 'Rok'
print("\n--- Ramka z listy list ---")
display(df_z_listy)

# Transponowanie (zamiana kolumn i wierszy)
df_transponowana = df_ze_slownika.T
print("\n--- Ramka transponowana ---")
display(df_transponowana)

# Wyświetlić pierwsze 10 wierszy
print("--- Pierwsze 10 wierszy ---")
display(df.head(10))

# Wyświetlić ostatnie 10 wierszy
print("\n--- Ostatnie 10 wierszy ---")
display(df.tail(10))

# Informacja o ramce danych
print("\n--- Informacje o ramce danych ---")
df.info()

# Ile wierszy i kolumn
wiersze, kolumny = df.shape
print(f"\nRamka ma {wiersze} wierszy oraz {kolumny} kolumn.")

# Statystyki kolumn liczbowych
print("\n--- Statystyki kolumn liczbowych ---")
display(df.describe())

# Statystyki kolumn kategoryzowanych
print("\n--- Statystyki kolumn kategoryzowanych ---")
display(df.describe(include=['object']))
```

```
# Usunięcie brakujących wartości
df = df.dropna()
print(f"\nRozmiar ramki po usunięciu brakujących wartości: {df.shape}")

# Wybór wierszy i kolumn po nazwach i indeksach
display(df.loc[0:5, ['location_name', 'val']])
display(df.iloc[0:5, [0, 2]])

# Wybór wierszy pod warunkiem określonej wartości
display(df[df['year_id'] == '2019'].head())

# Wybór wierszy przy spełnieniu kilku warunków
display(df[(df['year_id'] == '2019') & (df['val'] > 0.1)].head())

# Wiersze, które zawierają w kolumnie określone słowo
display(df[df['location_name'].str.contains('Poland', na=False)].head())

# Wiersze, które nie zawierają określonego słowa
display(df[~df['location_name'].str.contains('Poland', na=False)].head())

# Utworzenie nowej kolumny na podstawie istniejącej
df['val_procent'] = df['val'] * 100

# Usunięcie kolumny
df = df.drop(columns=['val_procent'])

# Zmiana nazwy kolumny
df = df.rename(columns={'val': 'Wartosc'})

# Zachowanie ramki jako plik csv
df.to_csv('wariant5_przetworzone.csv', index=False)

# Średnia, maksymalna i minimalna wartość z jednej kolumny
print("Średnia:", df['Wartosc'].mean())
print("Maksimum:", df['Wartosc'].max())
print("Minimum:", df['Wartosc'].min())

# Wyświetlenie liczby wierszy
print("Liczba wierszy:", len(df))

# Wartości unikatowe w kolumnie
print("Unikalne lata:", df['year_id'].unique())

# Liczby rekordów odpowiadających wartościom
print(df['location_name'].value_counts().head())

# Sortowanie wierszy malejąco i rosnąco
display(df.sort_values(by='Wartosc', ascending=False).head())
display(df.sort_values(by='Wartosc', ascending=True).head())

# 10 największych i najmniejszych wartości
display(df.nlargest(10, 'Wartosc'))
display(df.nsmallest(10, 'Wartosc'))

# 10 największych wartości pod warunkiem określonej wartości innej kolumny
dane_2019 = df[df['year_id'] == '2019']
display(dane_2019.nlargest(10, 'Wartosc'))
# Grupowanie (MultiIndex) i uśrednienie wartości wszystkich kolumn
kolumny_liczbowe = df.select_dtypes(include=['number']).columns
```

```
df_grupowane = df.groupby(['location_name', 'year_id'])[kolumny_liczbowe].mean()
display(df_grupowane.head())

# Grupowanie - średnia, liczba, mediana
display(df.groupby('location_name').agg({
    'Wartosc': 'mean',
    'year_id': 'count',
    'upper': 'median'
}).head())

# Nazwy kolumn indeksu złożonego
print("Nazwy w MultiIndex:", df_grupowane.index.names)

# Sortowanie kolumny indeksu złożonego
display(df_grupowane.sort_index(level='location_name', ascending=False).head())

# Tabela przestawna (pivot table)
tabela_przestawna = pd.pivot_table(df, values='Wartosc', index=['location_name']

# Wyświetlenie indeksów i kolumn tabeli przestawnej
print("Indeksy tabeli:", tabela_przestawna.index)
print("Kolumny tabeli:", tabela_przestawna.columns)

# Indeks złożony tabeli przestawnej
tabela_zlozona = pd.pivot_table(df, values='Wartosc', index=['location_name', 's
display(tabela_zlozona.head())

# Wykres na podstawie tabeli przestawnej
tabela_przestawna.head(5).T.plot(figsize=(10, 6), title="Wartość wskaźnika w cza
plt.show()


# Narysować histogram na podstawie wartości kolumny (tylko dla Prevalence)
df_prevalence = df[df['measure_name'] == 'Prevalence']
df_prevalence['Wartosc'].plot(kind='hist', bins=20, title="Histogram: Rozkład Wa
plt.show()

# Metody merge i concat (łączenie)
df_czesc1 = df.head(100)
df_czesc2 = df.tail(100)
df_concat = pd.concat([df_czesc1, df_czesc2])

df_slownik = pd.DataFrame({'location_name': ['Poland', 'Germany'], 'Kod_kraju':
df_merge = pd.merge(df, df_slownik, on='location_name', how='inner')
display(df_merge.head())

# Nowe kolumny - operacje matematyczne
df['Roznica'] = df['upper'] - df['lower']

# Nowe kolumny - funkcja lambda
df['Kategoria'] = df['Wartosc'].apply(lambda x: 'Wysoka' if x > 0.2 else 'Niska'

# Praca z dużymi plikami (chunksize)
print("Przetwarzanie pliku partiami (chunksize):")
for paczka in pd.read_csv(nazwa_pliku, chunksize=50000):
    print(f"Rozmiar pobranej paczki: {paczka.shape}")
    break
```

```
--- Ramka ze słownika ---
```

	Kraj	Odsetek_palacych	Rok
0	Polska	15.2	2019
1	Niemcy	12.1	2019
2	Czechy	14.5	2019

```
--- Pomyślnie wczytano dane z pliku CSV ---

--- Ramka z listy list ---
```

	Kraj	Odsetek_palacych	Rok
0	Francja	13.0	2019
1	Hiszpania	16.2	2019
2	Włochy	14.8	2019

```
--- Ramka transponowana ---
```

	0	1	2
Kraj	Polska	Niemcy	Czechy
Odsetek_palacych	15.2	12.1	14.5
Rok	2019	2019	2019

```
--- Pierwsze 10 wierszy ---
```

	measure_name	location_id	location_name	sex_id	sex_name	age_group_id	age_grou
0	Prevalence	1	Global	1	Male	149	
1	Prevalence	1	Global	2	Female	149	
2	Prevalence	4	Southeast Asia, East Asia, and Oceania	1	Male	149	
3	Prevalence	4	Southeast Asia, East Asia, and Oceania	2	Female	149	
4	Prevalence	6	China	1	Male	149	
5	Prevalence	6	China	2	Female	149	
6	Prevalence	7	Democratic People's Republic of Korea	1	Male	149	
7	Prevalence	7	Democratic People's Republic of Korea	2	Female	149	
8	Prevalence	8	Taiwan (Province of China)	1	Male	149	
9	Prevalence	8	Taiwan (Province of China)	2	Female	149	



```
--- Ostatnie 10 wierszy ---
```

	measure_name	location_id	location_name	sex_id	sex_name	age_group_id	age_g
1262	Percent Change in Prevalence	413	Tokelau	1	Male	149	
1263	Percent Change in Prevalence	413	Tokelau	2	Female	149	
1264	Percent Change in Prevalence	416	Tuvalu	1	Male	149	
1265	Percent Change in Prevalence	416	Tuvalu	2	Female	149	
1266	Percent Change in Prevalence	422	United States Virgin Islands	1	Male	149	
1267	Percent Change in Prevalence	422	United States Virgin Islands	2	Female	149	
1268	Percent Change in Prevalence	435	South Sudan	1	Male	149	
1269	Percent Change in Prevalence	435	South Sudan	2	Female	149	
1270	Percent Change in Prevalence	522	Sudan	1	Male	149	
1271	Percent Change in Prevalence	522	Sudan	2	Female	149	

```
--- Informacje o ramce danych ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1272 entries, 0 to 1271
Data columns (total 11 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   measure_name    1272 non-null   object
 1   location_id     1272 non-null   int64
 2   location_name   1272 non-null   object
 3   sex_id          1272 non-null   int64
 4   sex_name        1272 non-null   object
 5   age_group_id    1272 non-null   int64
 6   age_group_name  1272 non-null   object
 7   year_id         1272 non-null   object
 8   val             1272 non-null   float64
 9   upper           1272 non-null   float64
 10  lower           1272 non-null   float64
dtypes: float64(3), int64(3), object(5)
memory usage: 109.4+ KB

Ramka ma 1272 wierszy oraz 11 kolumn.

--- Statystyki kolumn liczbowych ---
```

	location_id	sex_id	age_group_id	val	upper	lowe
count	1272.000000	1272.000000	1272.0	1.272000e+03	1.272000e+03	1.272000e+03
mean	134.287736	1.500000	149.0	3.645334e+05	4.033136e+05	3.301779e+05
std	97.404833	0.500197	0.0	4.069578e+06	4.268955e+06	3.887326e+06
min	1.000000	1.000000	149.0	-7.588282e-01	-6.736115e-01	-8.420966e-01
25%	61.750000	1.000000	149.0	2.111960e-02	1.288786e-01	-2.788391e-01
50%	121.500000	1.500000	149.0	2.258776e-01	4.263123e-01	1.246196e-01
75%	180.250000	2.000000	149.0	1.181078e+04	1.657995e+04	8.223125e+03
max	522.000000	2.000000	149.0	1.249957e+08	1.295466e+08	1.208124e+08

```
--- Statystyki kolumn kategoryzowanych ---
```

	measure_name	location_name	sex_name	age_group_name	year_id
count	1272	1272	1272	1272	1272
unique	3	212	2	1	2
top	Prevalence	Global	Male	15 to 24	2019
freq	424	6	636	1272	848

```
Rozmiar ramki po usunięciu brakujących wartości: (1272, 11)
```

	location_name	val
0	Global	0.201031
1	Global	0.049511
2	Southeast Asia, East Asia, and Oceania	0.320305
3	Southeast Asia, East Asia, and Oceania	0.026897
4	China	0.301431
5	China	0.021667

	measure_name	location_name
0	Prevalence	Global
1	Prevalence	Global
2	Prevalence	Southeast Asia, East Asia, and Oceania
3	Prevalence	Southeast Asia, East Asia, and Oceania
4	Prevalence	China

	measure_name	location_id	location_name	sex_id	sex_name	age_group_id	age_grou
0	Prevalence	1	Global	1	Male	149	
1	Prevalence	1	Global	2	Female	149	
2	Prevalence	4	Southeast Asia, East Asia, and Oceania	1	Male	149	
3	Prevalence	4	Southeast Asia, East Asia, and Oceania	2	Female	149	
4	Prevalence	6	China	1	Male	149	

	measure_name	location_id	location_name	sex_id	sex_name	age_group_id	age_grou
0	Prevalence	1	Global	1	Male	149	
2	Prevalence	4	Southeast Asia, East Asia, and Oceania	1	Male	149	
4	Prevalence	6	China	1	Male	149	
6	Prevalence	7	Democratic People's Republic of Korea	1	Male	149	
8	Prevalence	8	Taiwan (Province of China)	1	Male	149	

	measure_name	location_id	location_name	sex_id	sex_name	age_group_id	age_gro
86	Prevalence	51	Poland	1	Male	149	
87	Prevalence	51	Poland	2	Female	149	
510	Number of Smokers	51	Poland	1	Male	149	
511	Number of Smokers	51	Poland	2	Female	149	
934	Percent Change in Prevalence	51	Poland	1	Male	149	

	measure_name	location_id	location_name	sex_id	sex_name	age_group_id	age_grou
0	Prevalence	1	Global	1	Male	149	
1	Prevalence	1	Global	2	Female	149	
2	Prevalence	4	Southeast Asia, East Asia, and Oceania	1	Male	149	
3	Prevalence	4	Southeast Asia, East Asia, and Oceania	2	Female	149	
4	Prevalence	6	China	1	Male	149	

```
Średnia: 364533.3700294625
Maksimum: 124995683.5
Minimum: -0.758828209
Liczba wierszy: 1272
Unikalne lata: ['2019' '1990-2019']
location_name
Global                  6
Saudi Arabia            6
Tunisia                 6
Turkey                  6
United Arab Emirates    6
Name: count, dtype: int64
```

	measure_name	location_id	location_name	sex_id	sex_name	age_group_id	age_gro
424	Number of Smokers	1	Global	1	Male	149	
426	Number of Smokers	4	Southeast Asia, East Asia, and Oceania	1	Male	149	
425	Number of Smokers	1	Global	2	Female	149	
428	Number of Smokers	6	China	1	Male	149	
702	Number of Smokers	158	South Asia	1	Male	149	

	measure_name	location_id	location_name	sex_id	sex_name	age_group_id	age_g
1083	Percent Change in Prevalence	135	Brazil	2	Female	149	
1082	Percent Change in Prevalence	135	Brazil	1	Male	149	
1005	Percent Change in Prevalence	90	Norway	2	Female	149	
969	Percent Change in Prevalence	71	Australia	2	Female	149	
1230	Percent Change in Prevalence	216	Senegal	1	Male	149	

	measure_name	location_id	location_name	sex_id	sex_name	age_group_id	age_gro
424	Number of Smokers	1	Global	1	Male	149	
426	Number of Smokers	4	Southeast Asia, East Asia, and Oceania	1	Male	149	
425	Number of Smokers	1	Global	2	Female	149	
428	Number of Smokers	6	China	1	Male	149	
702	Number of Smokers	158	South Asia	1	Male	149	
710	Number of Smokers	163	India	1	Male	149	
534	Number of Smokers	64	High-income	1	Male	149	
662	Number of Smokers	137	North Africa and Middle East	1	Male	149	
535	Number of Smokers	64	High-income	2	Female	149	
716	Number of Smokers	166	Sub-Saharan Africa	1	Male	149	

	measure_name	location_id	location_name	sex_id	sex_name	age_group_id	age_g
1083	Percent Change in Prevalence	135	Brazil	2	Female	149	
1082	Percent Change in Prevalence	135	Brazil	1	Male	149	
1005	Percent Change in Prevalence	90	Norway	2	Female	149	
969	Percent Change in Prevalence	71	Australia	2	Female	149	
1230	Percent Change in Prevalence	216	Senegal	1	Male	149	
991	Percent Change in Prevalence	83	Iceland	2	Female	149	
1027	Percent Change in Prevalence	103	Latin America and Caribbean	2	Female	149	
1226	Percent Change in Prevalence	214	Nigeria	1	Male	149	
1064	Percent Change in Prevalence	125	Colombia	1	Male	149	
1004	Percent Change in Prevalence	90	Norway	1	Male	149	

	measure_name	location_id	location_name	sex_id	sex_name	age_group_id	age_gro
424	Number of Smokers	1	Global	1	Male	149	
426	Number of Smokers	4	Southeast Asia, East Asia, and Oceania	1	Male	149	
425	Number of Smokers	1	Global	2	Female	149	
428	Number of Smokers	6	China	1	Male	149	
702	Number of Smokers	158	South Asia	1	Male	149	
710	Number of Smokers	163	India	1	Male	149	
534	Number of Smokers	64	High-income	1	Male	149	
662	Number of Smokers	137	North Africa and Middle East	1	Male	149	
535	Number of Smokers	64	High-income	2	Female	149	
716	Number of Smokers	166	Sub-Saharan Africa	1	Male	149	

		location_id	sex_id	age_group_id	Wartosc	upper
location_name	**year_id**					
Afghanistan	**1990-2019**	160.0	1.5	149.0	2.233005	4.514392
	2019	160.0	1.5	149.0	181766.166798	231175.223577
Albania	**1990-2019**	43.0	1.5	149.0	0.845445	1.913974
	2019	43.0	1.5	149.0	29698.325518	34582.314961
Algeria	**1990-2019**	139.0	1.5	149.0	-0.061563	0.537529

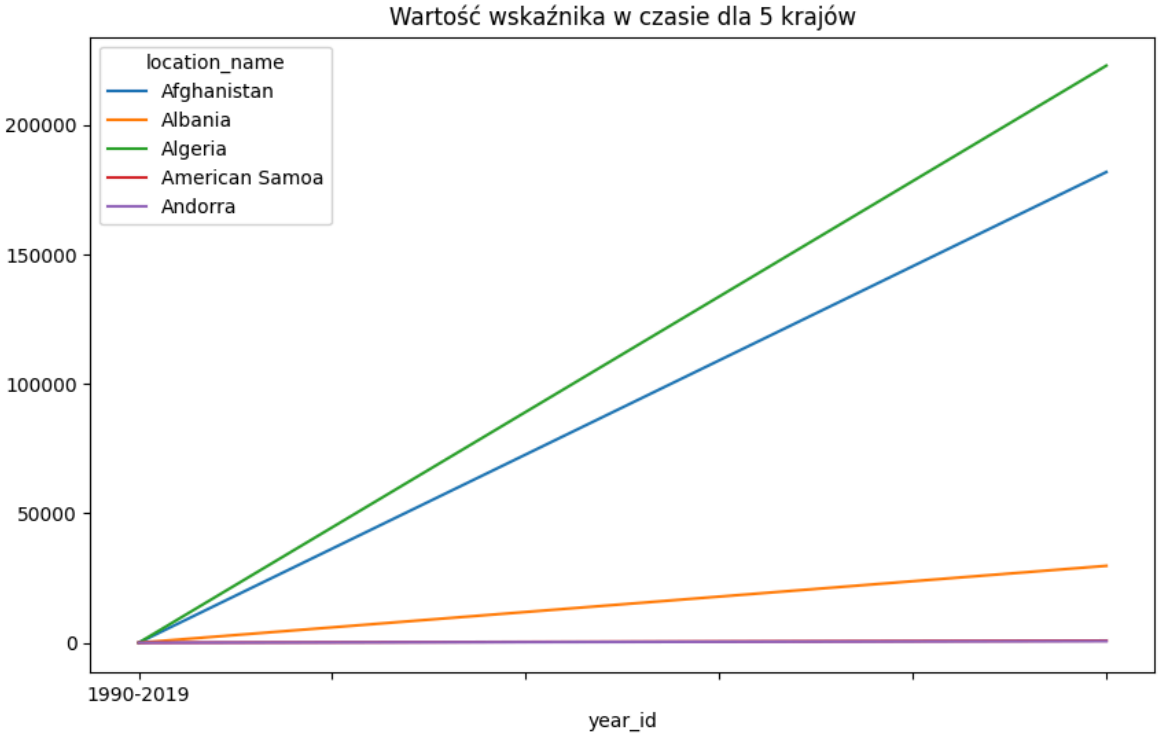
location_name	Wartosc	year_id	upper
Afghanistan	121178.188867	6	4.514392
Albania	19799.165494	6	1.913974
Algeria	148576.461649	6	0.537529
American Samoa	548.214303	6	0.602564
Andorra	405.820017	6	0.386797

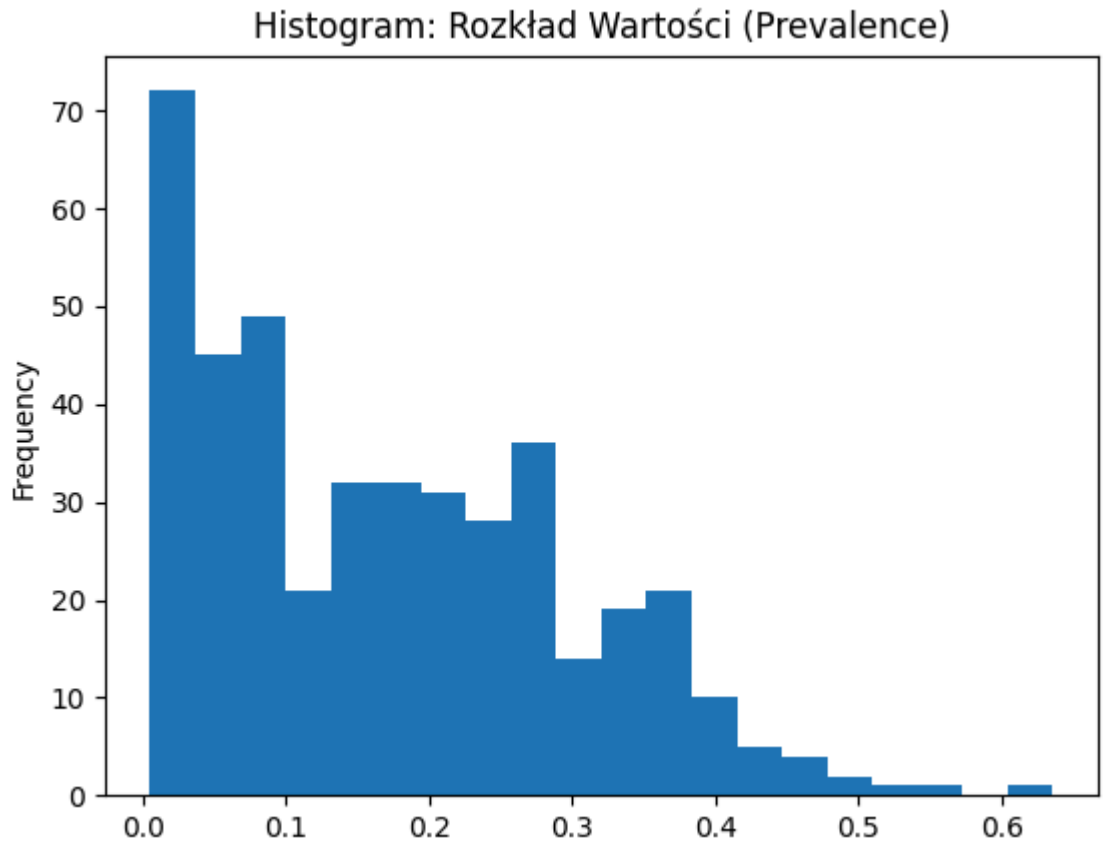
```
Nazwy w MultiIndex: ['location_name', 'year_id']
```

location_name	year_id	location_id	sex_id	age_group_id	Wartosc	upper
Zimbabwe	**2019**	198.0	1.5	149.0	75625.349866	96025.834349
	1990-2019	198.0	1.5	149.0	1.034468	2.923965
Zambia	**2019**	191.0	1.5	149.0	110488.390937	141103.551228
	1990-2019	191.0	1.5	149.0	0.781116	2.136812
Yemen	**2019**	157.0	1.5	149.0	181815.709562	234407.796329

```
Indeksy tabeli: Index(['Afghanistan', 'Albania', 'Algeria', 'American Samoa', 'An
dorra',
       'Angola', 'Antigua and Barbuda', 'Argentina', 'Armenia', 'Australia',
       ...
       'United States Virgin Islands', 'United States of America', 'Uruguay',
       'Uzbekistan', 'Vanuatu', 'Venezuela (Bolivarian Republic of)',
       'Viet Nam', 'Yemen', 'Zambia', 'Zimbabwe'],
      dtype='object', name='location_name', length=212)
Kolumny tabeli: Index(['1990-2019', '2019'], dtype='object', name='year_id')
```

location_name	sex_name	1990-2019	2019
Afghanistan	**Female**	1.992220	49585.904205
	Male	2.473790	313946.429391
Albania	**Female**	1.040573	12200.927966
	Male	0.650316	47195.723070
Algeria	**Female**	-0.136860	25322.627340


Wartość wskaźnika w czasie dla 5 krajów
location_name
Afghanistan
Albania
Algeria
American Samoa
Andorra
200000
150000
100000
50000
0
1990-2019
year_id


Histogram: Rozkład Wartości (Prevalence)
Frequency
70
60
50
40
30
20
10
0
0.0
0.1
0.2
0.3
0.4
0.5
0.6

	measure_name	location_id	location_name	sex_id	sex_name	age_group_id	age_grou
0	Prevalence	51	Poland	1	Male	149	
1	Prevalence	51	Poland	2	Female	149	
2	Prevalence	81	Germany	1	Male	149	
3	Prevalence	81	Germany	2	Female	149	
4	Number of Smokers	51	Poland	1	Male	149	

```
Przetwarzanie pliku partiami (chunksize):
Rozmiar pobranej paczki: (1272, 11)
```